

Demo Circuit for Over-Voltage Protection

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Over-voltage protection circuits are used to protect voltage-sensitive loads. Voltage transients may occur due to a number of reasons such as transformer switching, load switching, and short/open circuit in rectifier and regulator circuit. Such transients can affect proper functioning of an electronic circuit or even damage it. Hence it is necessary to use an over-voltage protection circuit to protect expensive loads against all the sources

of voltage transients.

In electronics engineering, where over-voltage protection experiment is included in the syllabus, the present circuit can be used to very effectively demonstrate the effect to students.

Circuit and working

Fig. 1 shows the demo circuit for over-voltage protection. It is built around



PARTS LIST

Semiconductors:

- IC1 - 7810, 10V regulator
- SCR1 - 2P4M SCR
- T1 - BC548 npn transistor
- ZD1 - 11V zener diode
- D1-D4 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R3 - 10-kilo-ohm
- R2 - 2.2-kilo-ohm
- R4 - 50-ohm, 5W
- R5 - 2.2-mega-ohm
- R6 - 6.8-kilo-ohm
- VR1 - 1-kilo-ohm potmeter

Capacitors:

- C1 - 470 μ F, 40V electrolytic
- C2 - 100 μ F, 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 15V, 1A secondary transformer
- CON1 - 2-pin connector for 230V AC mains
- CON2 - 2-pin connector for 6V bulb
- CON3 - 2-pin connector for digital multimeter
- S1 - On/off switch
- F1 - 500mA fuse
- 6V bulb

TABLE I
Output Voltages
Corresponding to Different
Settings of VR1

Resistance of VR1	Output	Fuse condition
1 kilo-ohm	10.30V	Ok
820 ohms	10.94V	Ok
680 ohms	11.60V	Blown

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	10V when S1 'on'
TP4	0.7V (when VR1 = 1kilo-ohm) 1.5V (when VR1 = 680 ohms)
Across TP2 and TP3	230V AC when S1 'on'

a rectifier comprising four 1N4007 diodes (D1 through D4), 10V voltage regulator IC 7810 (IC1), SCR 2P4M (SCR1), transistor BC548 (T1) and a few other components. SCR1 is used as a protective component.

If voltage exceeds beyond the withstanding voltage capacity of the device that needs to be protected (6V bulb here), the circuit disconnects the device from supply. To demonstrate this, potmeter VR1 connected across regulator IC1 is used to increase the voltage at the output of regulator IC1. When the output voltage of IC1 increases, voltage at the base of transistor T1 also increases, which triggers SCR1 through resistor R6.

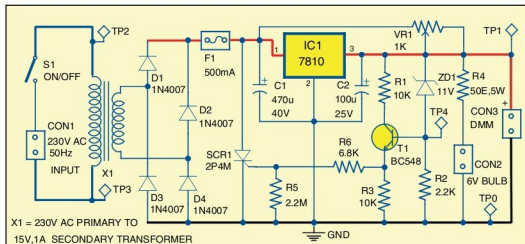


Fig. 1: Demo circuit for over-voltage protection

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